

MASS-APP-014-V16 — Intelligence Quality, Evaluation & Model Governance

Document Control

Field | Value

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1. Purpose

V16 measures whether AI-assisted outputs, recommendations, summaries, classifications, agents, plugins, and synthesized context are useful, accurate, grounded, safe, traceable, timely, and appropriate. It provides evaluation evidence and release advice without allowing AI to certify itself or authorize deployment.

2. Permanent Architecture

V16 owns evaluation programs, registered datasets, scenarios, test cases, acceptance criteria, evaluation runs, observed metrics, evidence, quality findings, comparative regressions, human judgments, waivers, release advisories, configuration history, and executive quality briefings. ENG-009 owns orchestration, ENG-011 observability, ENG-024 analytics, ENG-027 lineage, and source applications own evaluated outputs.

The permanent model separates observed measurements, deterministic checks, human judgment, and advisory synthesis. Each result identifies evaluator version, subject version, dataset version, evidence, limitations, and reviewer.

3. Role Mapping

Role | Baseline Mapping | Authority

Evaluation Viewer | Viewer specialization | Read authorized plans and findings

Evaluation Contributor | Contributor specialization | Register cases and execute approved runs

Quality Steward | Steward specialization | Govern criteria, datasets, findings, and comparisons

Human Evaluator | Contributor extension | Record independent qualitative judgment

Release Advisory Approver | Administrator specialization | Issue advisory or waiver decision

Evaluation Administrator | Administrator specialization | Configure gateways, roles, and retention

Evaluated agents/models cannot review themselves. Run owners cannot approve their own release advisory or waiver.

4. Platform Consumption Map

Source | Consumption | Boundary

APP-014 V01–V15 | Evaluated outputs and configuration references | Prior volumes retain ownership

APP-015 / APP-017 when available | Plugin and AI workforce contracts | No ownership before availability

ENG-003/004/005 | Identity, authorization, events | Platform-owned

ENG-007/009/011 | Knowledge, AI execution, observations | Platform-owned

ENG-015/024/027 | APIs, metrics, lineage | Capability-owned

External providers | Approved model/evaluation references | Accessed through gateways only

Publishes evaluation.run.completed, evaluation.finding.raised, evaluation.regression.detected, evaluation.waiver.decided, evaluation.release-advisory.issued.

5. Gateway Inventory

Gateway | Contract | Failure Behavior

EvaluatedArtifactGateway | APP-014 output/configuration references | Mark case unexecutable if version unavailable

PluginContractGateway | APP-015 contracts when available | Skip unavailable optional subject

WorkforceContractGateway | APP-017 contracts when available | No assumed contract before publication

AIOrchestrationGateway | ENG-009 controlled execution | Fail run without fabricated output

KnowledgeGateway | ENG-007 grounding corpus | Mark grounding incomplete

ObservabilityGateway | ENG-011 trace evidence | Preserve missing telemetry finding

AnalyticsGateway | ENG-024 metrics | Preserve raw observations if aggregation fails

LineageGateway | ENG-027 citation validation | Block release advisory when mandatory lineage fails

ExternalEvaluationGateway | Approved providers | Isolate provider result as external evidence

EventGateway / AuditGateway | ENG-005/011 | Outbox and retry-safe audit

6. Evaluation Lifecycle

Program Drafted → Dataset Registered → Cases Approved → Run Scheduled → Outputs Captured → Measures Calculated → Human Reviewed → Findings Issued → Compared → Release Advisory / Waiver → Preserved

Datasets and acceptance criteria are versioned. A completed run is immutable. A rerun creates a new run linked to its predecessor.

7. Evaluation Model

Test cases define input, context, expected properties, prohibited properties, acceptance criteria, severity, and required reviewer. Expected output may be exact, rubric-based, reference-based, constraint-based, or human-judgment based.

Measures cover accuracy, relevance, completeness, grounding, consistency, timeliness, usefulness, citation validity, unsupported claims, safety, policy conformance, and observed bias/disparate impact. Measurements remain observations; human reviewers determine organizational acceptability.

Comparisons preserve baseline and candidate versions, matched datasets, metric deltas, changed failures, confidence, and limitations. Threshold breaches create warnings and release recommendations, not deployment decisions.

8. Exceptions, Waivers, and Briefings

Exceptions record failed criteria and affected use. Waivers require risk, compensating controls, scope, expiration, and an authorized independent decision. Failed results remain visible.

NOVA evaluation advisory and POPS historical performance context execute through ENG-009 and remain cited advisory evidence. Executive briefings distinguish measurements, human judgments, unresolved findings, waivers, and recommendations.

9. Data Model

Entity | Purpose | Immutable Condition

EvaluationProgram | Governed evaluation scope | After activation

EvaluationDataset | Dataset identity/version | After validation

EvaluationScenario | Organizational scenario | After approval

EvaluationTestCase | Input and criteria | After approval

EvaluationSubject | Model/prompt/agent/plugin reference | Version record immutable

EvaluationRun | Executed evaluation | After completion

EvaluationObservation | Raw observed metric | Always after capture

HumanEvaluation | Reviewer judgment | Always

QualityFinding | Failed or warning condition | After review

RegressionComparison | Baseline/candidate comparison | After issuance

EvaluationWaiver | Human exception decision | Always

ReleaseAdvisory | Advisory release recommendation | After issuance

ExecutiveQualityBriefing | Cited quality summary | After issuance
EvaluationAdvisoryRequest | NOVA/POPS request | After completion
EvaluationOutbox | Reliable event | After publication

10. API Contracts

Method | Path | Purpose | Role

GET/POST | /evaluation/programs | Browse/create programs | Viewer/Steward
POST | /evaluation/datasets | Register dataset version | Quality Steward
POST | /evaluation/test-cases | Create test case | Evaluation Contributor
POST | /evaluation/runs | Start approved run | Evaluation Contributor
GET | /evaluation/runs/{id} | Read run and evidence | Evaluation Viewer
POST | /evaluation/runs/{id}/human-reviews | Record judgment | Human Evaluator
GET | /evaluation/findings | Browse findings | Evaluation Viewer
POST | /evaluation/comparisons | Compare candidate to baseline | Quality Steward
POST | /evaluation/waivers | Request waiver | Quality Steward
POST | /evaluation/waivers/{id}/decide | Decide waiver | Release Advisory Approver
POST | /evaluation/release-advisories/{id}/issue | Issue advisory | Release Advisory Approver
POST | /evaluation/briefings/{id}/issue | Issue briefing | Release Advisory Approver
POST | /evaluation/advisory-requests | Request NOVA/POPS advice | Quality Steward

11. Security and Integrity

Tenant RLS derives from `auth.jwt()`. Composite foreign keys protect tenant boundaries. Dataset versions, approved cases, subject versions, completed runs, observations, human judgments, reviewed findings, comparisons, waiver decisions, issued advisories/briefings, and completed advisory responses are immutable. Separation-of-duty triggers prohibit self-certification.

Implementation-grade schema is in `MASS-APP-014-V16_Migration_Reference.sql`.

12. V1 Implementation

V1 supports registered evaluation programs and datasets, scenario/test-case stewardship, controlled runs through ENG-009, deterministic and rubric measures, citation checks, human review, regression comparison, findings, time-bounded waivers, release advisories, and executive quality briefings.

13. Future Evolution

Expanded adversarial testing, richer fairness evaluation, external benchmark providers, continuous regression monitoring, and additional subject types attach through existing gateway, subject, dataset, and review contracts. V16 never becomes autonomous certification authority.

14. Constitutional Boundary Statement

V16 measures, compares, explains, flags, and recommends. It does not alter evaluated outputs, suppress failures, certify itself, authorize deployment, override policy, or execute correction.